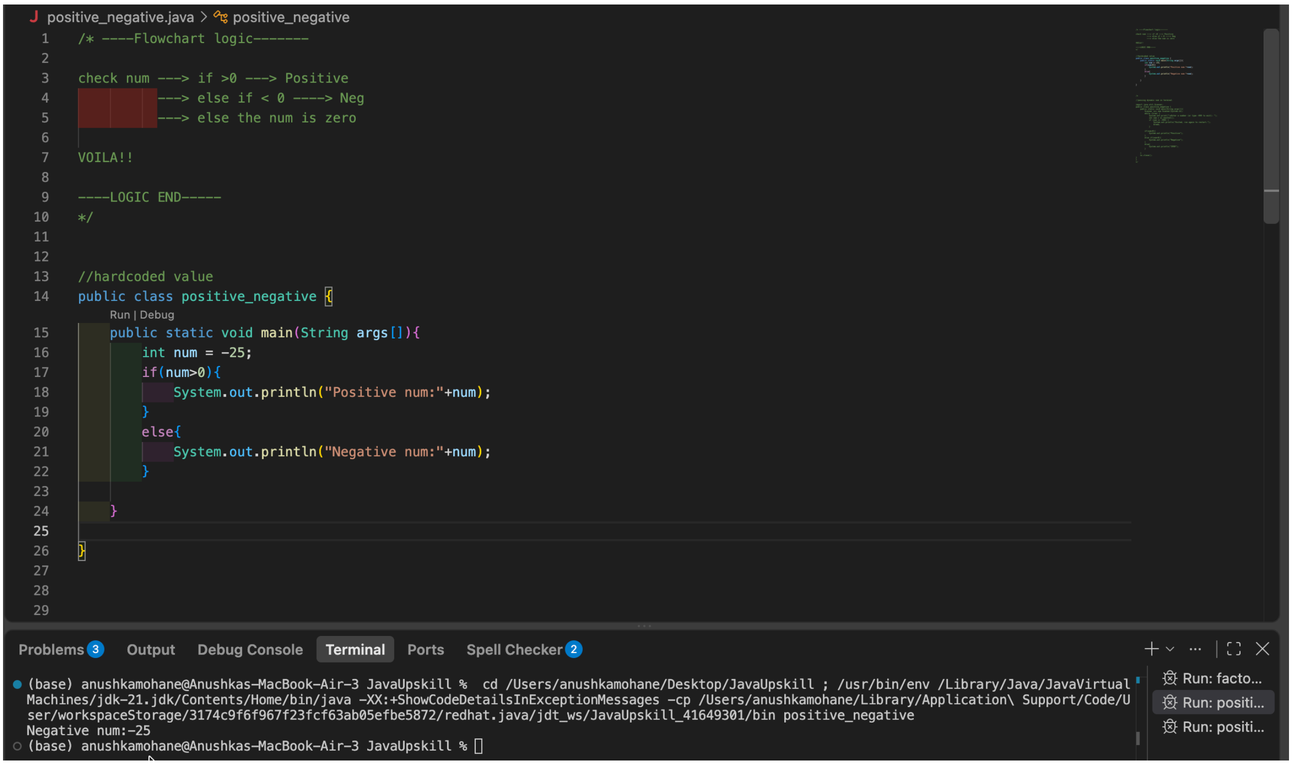
Anushka_Mohane_AC Lab 1

Answer to Q1 & Q2 together below:

The flowchart logic is at the top.
-> Built the first one with hardcoded value (img 1)
-> Coded the second one with dynamic value (img 2)

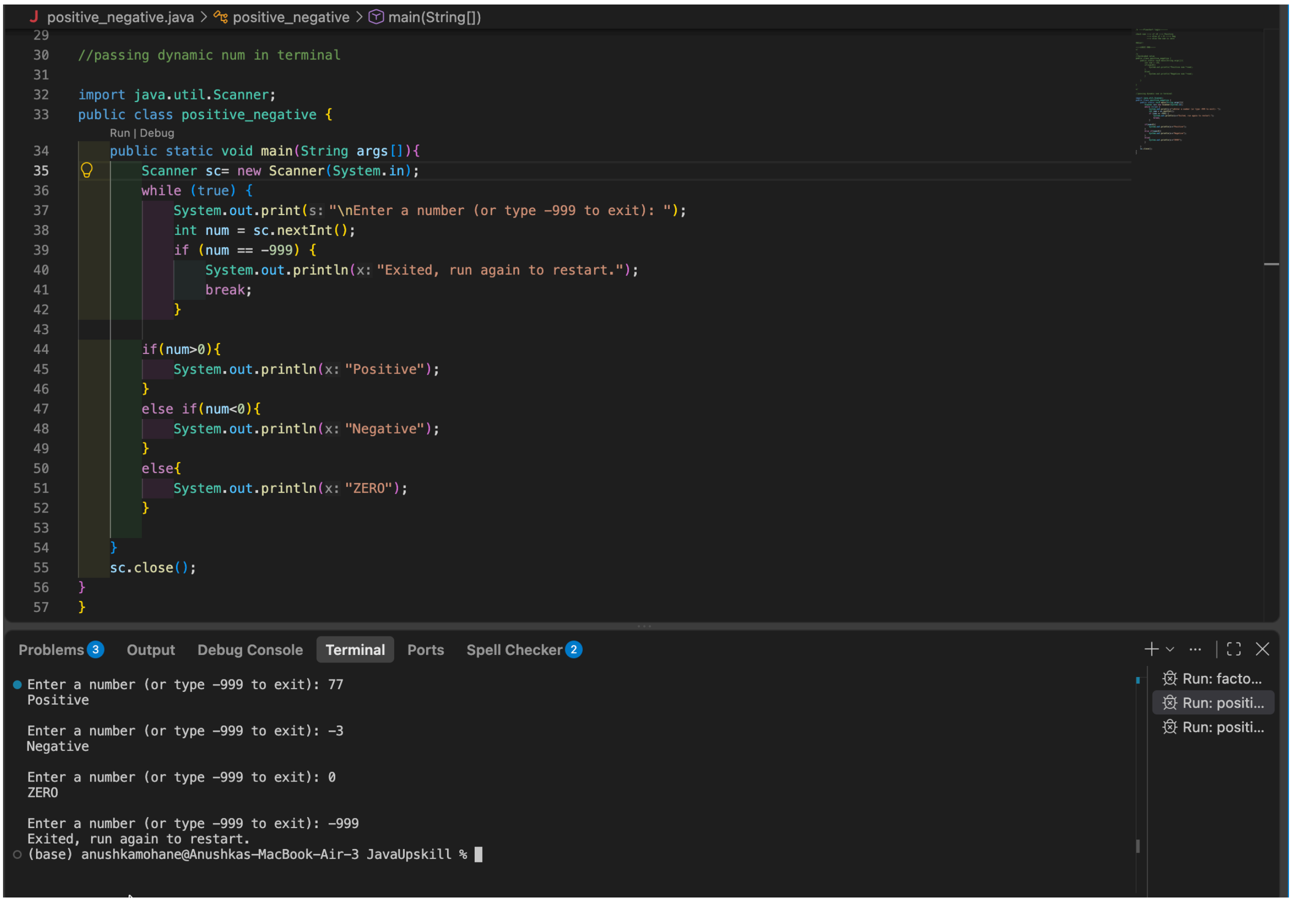


positive_negative.java > positive_negative

```java
/* ----Flowchart logic-------

check num ---> if >0 ---> Positive
          ---> else if < 0 ----> Neg
          ---> else the num is zero

VOILA!!

----LOGIC END-----
*/


//hardcoded value
public class positive_negative {
    public static void main(String args[]){
        int num = -25;
        if(num>0){
            System.out.println("Positive num:"+num);
        }
        else{
            System.out.println("Negative num:"+num);
        }

    }

}
```

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %  cd /Users/anushkamohane/Desktop/JavaUpskill ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin positive_negative
Negative num:-25
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

IMG 1 - Flowchart + Hardcoded number



positive_negative.java > positive_negative > main(String[])

```java
//passing dynamic num in terminal

import java.util.Scanner;
public class positive_negative {
    public static void main(String args[]){
        Scanner sc= new Scanner(System.in);
        while (true) {
            System.out.print(s: "\nEnter a number (or type -999 to exit): ");
            int num = sc.nextInt();
            if (num == -999) {
                System.out.println(x: "Exited, run again to restart.");
                break;
            }

            if(num>0){
                System.out.println(x: "Positive");
            }
            else if(num<0){
                System.out.println(x: "Negative");
            }
            else{
                System.out.println(x: "ZERO");
            }

        }
        sc.close();
    }
}
```

```
Enter a number (or type -999 to exit): 77
Positive

Enter a number (or type -999 to exit): -3
Negative

Enter a number (or type -999 to exit): 0
ZERO

Enter a number (or type -999 to exit): -999
Exited, run again to restart.
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

IMG 2 - Dynamic number

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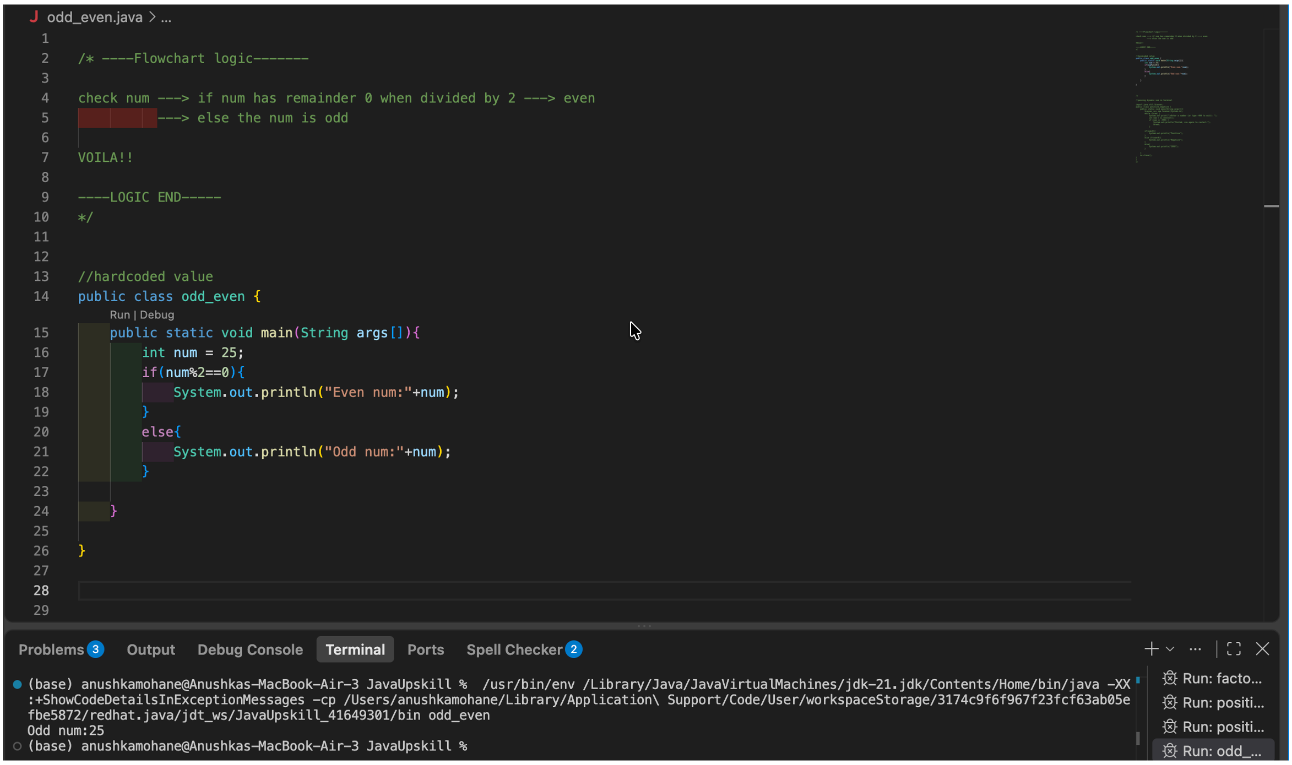
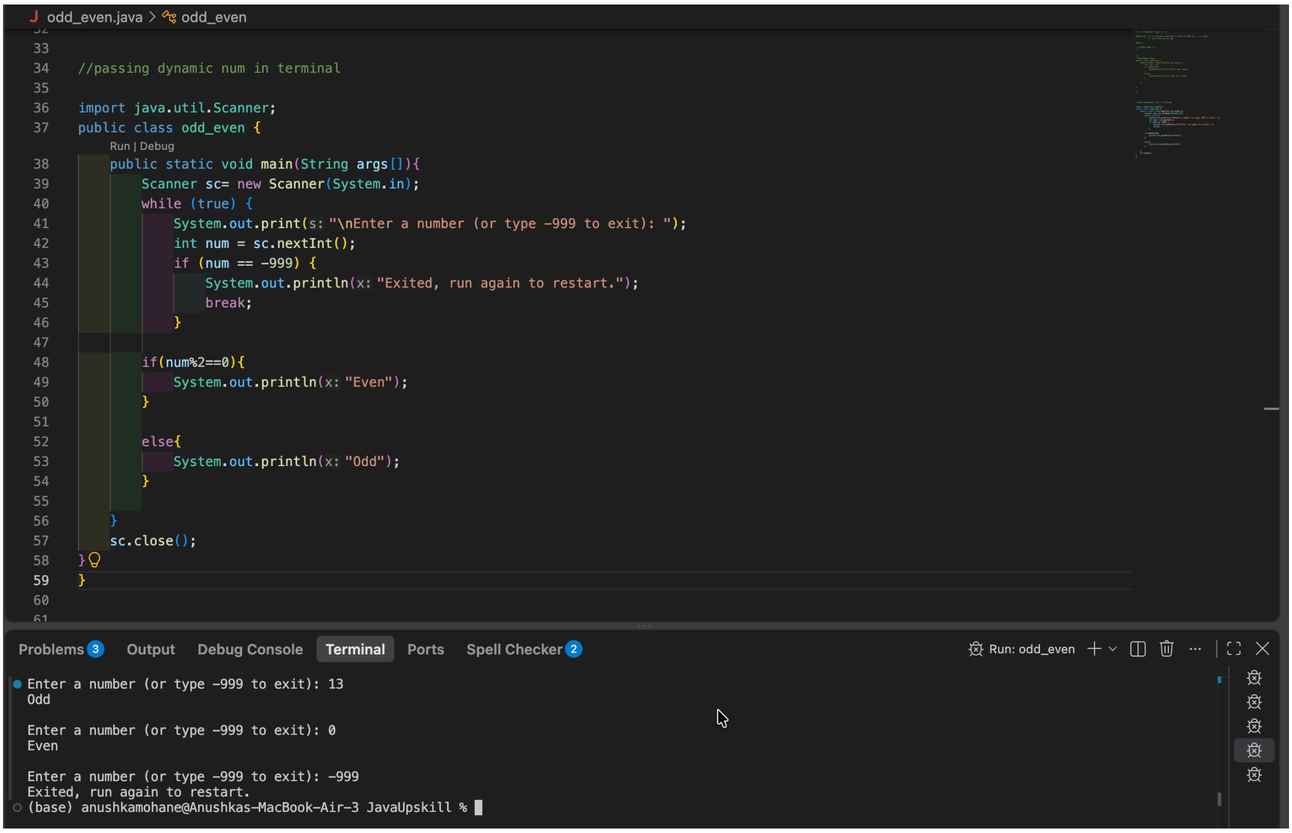
Answer to Q3:



IMG 3 - Flowchart + Hardcoded number



IMG 4 - Dynamic number

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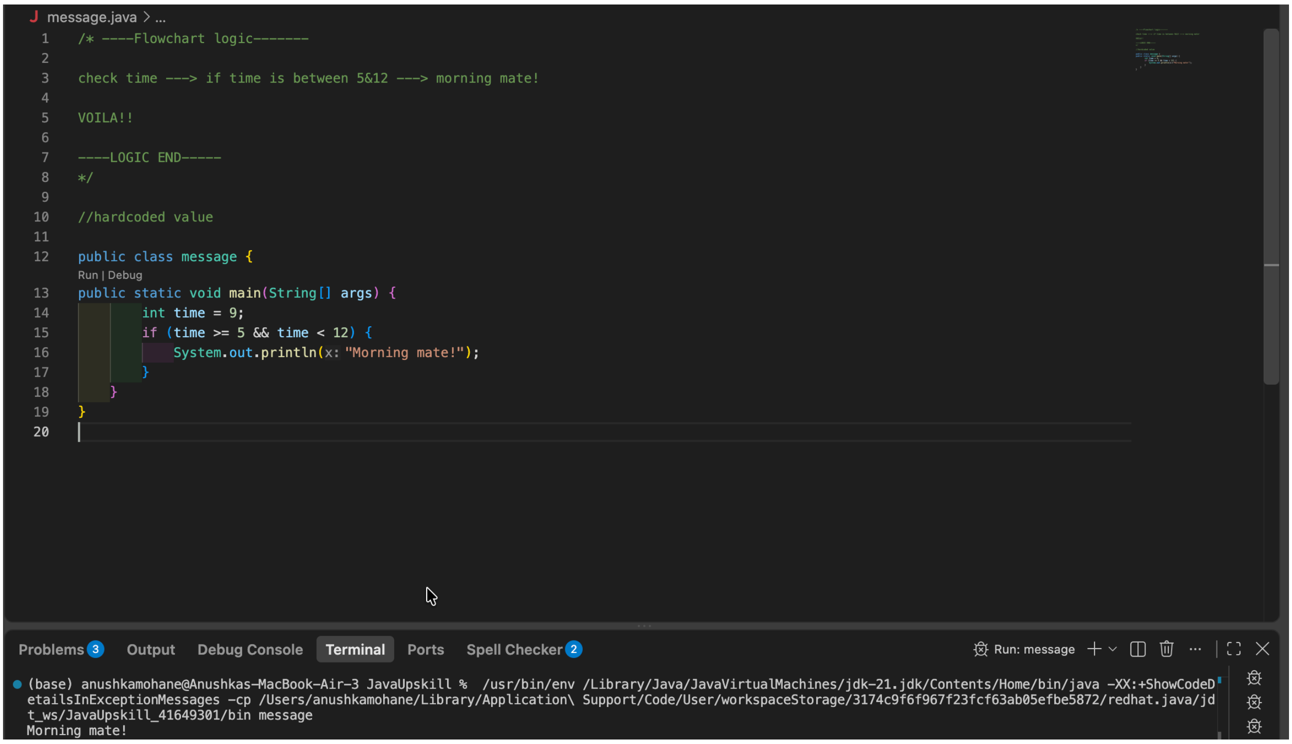
Answer to Q4:



IMG 5 - Flowchart + Hardcoded number

Answer to Q5 & Q6:

```
area_square_rectangle.java > ...
/* ----Flowchart logic-------

im using case statements to combine Q5 & Q6 into one program.

have predefined variables for calculations -> if user enters choice 1-> calculate area of square with formula of side square.
                                           -> if user enters choice 2-> calculate area of rectangle with formula of length*width.

VOILA!!

----LOGIC END-----
*/
```

Img 6: Flowchart Logic. I have used Case statement logic to combine Q5 & Q6

```java
area_square_rectangle.java > ...
import java.util.Scanner;
public class area_square_rectangle {
    Run | Debug
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);

        while (true) {
            System.out.println(x: "Enter 1 for area of square and 2 for area of rectangle (or type -999 to exit): ");
            int c = sc.nextInt();
            if (c == -999) {
                System.out.println(x: "Exited, run again to restart.");
                break;
            }


        double side = 5.0;
        double length = 8.0;
        double width = 4.0;

        switch (c) {
            case 1:
                double Ar_sq = side * side;
                System.out.println("Area of Square: " + Ar_sq);
                break;

            case 2:
                double Ar_rec = length * width;
                System.out.println("Area of Rectangle: " + Ar_rec);
                break;

            default:
                System.out.println(x: "Upar samjhaya hai sirf 1 ya 2 likhe, what is this behaviour lol");
                break;
        }

    }
    sc.close();
}
}
```

Img 7: Code

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %  /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin area_square_rectangle
Enter 1 for area of square and 2 for area of rectangle (or type -999 to exit):
1
Area of Square: 25.0
Enter 1 for area of square and 2 for area of rectangle (or type -999 to exit):
2
Area of Rectangle: 32.0
Enter 1 for area of square and 2 for area of rectangle (or type -999 to exit):
4
Upar samjhaya hai sirf 1 ya 2 likhe, what is this behaviour lol
Enter 1 for area of square and 2 for area of rectangle (or type -999 to exit):
```

Img 8: Output

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Answer to Q7:

largest.java > largest

```java
/* Flowchart logic:

If i want to code via basic if-else logic ==>
    compare a with b and c to check if a is greater than both
    -> else compare b & c to check if b is greater than c
    -> else c is largest

If i want to code via ternary logic this is how i'll do==>
    Compare a and b to check if a>=b ---> if true ---> compare a & c and check which is largest
                                    ---> if false ---> compare b & c and check which is largest

Voilaaaa !!!
*/
```

Img 9: Flowchart logic - for both if-else and the ternary operator approaches

```java
public class largest {
    public static void main(String[] args) {
        int a = 66;
        int b = 77;
        int c = 88;

        int largest;

        if (a >= b && a >= c) {
            largest = a;
        } else if (b >= c) {
            largest = b;
        } else {
            largest = c;
        }

        System.out.println("The largest number is: " + largest);
    }
}


//im trying to use ternary operator approach: quite easy to compute with less lines of code
```

Problems 9 Output Debug Console Terminal Ports Spell Checker 8 Run: largest

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %  cd /Users/anushkamohane/Desktop/JavaUpskill ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin largest
The largest number is: 88
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

Img 10: Code+output using if-else approach

largest.java > ...

```java

//im trying to use ternary operator approach: quite easy to compute with less lines of code

public class largest {
    public static void main(String[] args) {
        int num1 = 66;
        int num2 = 77;
        int num3 = 96;

        int largest = (num1 >= num2) ? ((num1 >= num3) ? num1 : num3) : ((num2 >= num3) ? num2 : num3);

        System.out.println("The largest number is: " + largest);
    }

}
```

Problems 9 Output Debug Console Terminal Ports Spell Checker 8 Run: largest

```
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %  cd /Users/anushkamohane/Desktop/JavaUpskill ; /usr/bin/env /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home/bin/java -XX:+ShowCodeDetailsInExceptionMessages -cp /Users/anushkamohane/Library/Application\ Support/Code/User/workspaceStorage/3174c9f6f967f23fcf63ab05efbe5872/redhat.java/jdt_ws/JavaUpskill_41649301/bin largest
The largest number is: 96
(base) anushkamohane@Anushkas-MacBook-Air-3 JavaUpskill %
```

Img 11: Code+output using ternary operator approach